2021 9th International Japan-Africa Conference on Electronics, Communications, and Computations (JAC-ECC) | 978-1-6654-8292-9/21/$31.00 ©2021 IEEE | DOI: 10.1109/JAC-ECC54461.2021.9691437

Control of Robot Arm Based on EOG Signals

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***Abstract*— This paper presents an Electrooculography (EOG) acquisition system for Human-Computer Interaction (HCI) based on Arduino microcontroller. The proposed system is used to acquire vertical and horizontal eye movements for control of wireless robot arm. The NRF24L01 - 2.4G wireless transceiver module is used to transfer data from master to slave circuits. The L298N driver circuit is mounted on a special robotic arm. The EOG is a signal produced in both positive and negative directions by eye movement. The EOG acquisition system is interfaced with Arduino Uno and Arduino Nano microcontrollers. Robotic arm movement is controlled by five basic eye movements in main directions. Five Ag-Agcl disposable electrodes are used to measure the potential difference for eye movement between cornea and retina. Different volunteers of different ages at different times have been treated with the proposed system to obtain data. For controlling the wireless robot arm, the Arduino microcontroller is used to serialize the EOG signals for calibration. The performance and accuracy of the presented system are really promising due to simplicity and reliability. This trend can help the disabled patients in wheelchair regulation, rehabilitation aids and other applications by simply moving their eyes via the proposed system.**

Keywords — Human-Computer Interaction (HCI), Robot Arm, Electrooculography (EOG), Electroencephalogram (EEG), Arduino, Electrocardiogram (ECG)\, Electromyography (EMG)

I. INTRODUCTION

The HCI based on EOG is an interesting research area that has evolved in recent years. The HCI provides communication streams between individuals and external devices. Studies of HCI could significantly improve the quality of life of people with severe disabilities, particularly those with spinal cord injuries, joint deformities, muscle spasms, and Amyotrophic Lateral Sclerosis (ALS) [1]. There are four bio-signals covering all activities performed by humans that are EOG, (, EEG and ECG [2]. The patients may have difficulties reaching their goals, because they cannot voluntarily control their muscles, but they are still conscious and can control eye movements. There are several devices that can be used to record the movements of the human eye, but most of them have limitations [3].

The EOG signals are widely used for determining eye functions and for understanding eye movements associated with them [4]. Feature extraction and classification are the major components of an HCI system. As a result, several EOG signaling tasks based HCI systems were configured [5, 6 and 7]. To measure and analyze the eye movement directions, one of the most important biomedical signals used for this task is the EOG [8].

EOG signals are induced due to the potential difference between cornea and retina [9]. Typically, EOG signals have a differential capacitance of 0.05 to 3.5 mV and a frequency range from 0.1 to 20 Hz. The HCI depends on EOG signals to transfer information between humans and machines [10, 11]. Several HCI applications have been proposed based on EOG signals, such as electric wheelchair control [12], mobile robot control [13] and hand-writing recognition [14-16]. In addition, wireless control can be performed using Bluetooth with microcontrollers [17].

In [18], the authors designed an EOG acquisition system by using INA 128P for amplifier with a bandwidth from 0.165Hz to 186 Hz. In [19], the authors designed and implemented an EOG signal acquisition system by using INA 128P and OP07P OP-AMP for filters with a bandwidth from 0.15Hz to 27 Hz. In [20], the authors designed and implemented an EOG signal acquisition system by using INA333 for the amplifier with active HPF and LPF and a bandwidth of 0.05Hz - 35Hz through the OPA2335.

In [21], the authors recorded a dataset for four patients during a variable range of visits from 2 to 10. Each visit had 3.22 ± 1.21 days and consisted of 5.57 ± 2.61 sessions recorded per day. The data was recorded based of EEG. In [22], the authors obtained the EOG data using the g.tec g.USBamp bio-signal amplifier. The data was bandpass filtered between 0 and 30 Hz.

In the proposed system in this paper, the EOG signal amplification and filtering circuit is built using AD620 and LM324N-IA. Arduino Uno and Arduino Nano microcontrollers are used for acquisition and transfer of the digital EOG signals to a PC by using NRF24L01-2.4G wireless transceiver module. MATLAB R2019.b and Arduino program are used on the PC to record and store signals from volunteers. The data is used to control a wireless EOG-based robot arm. The main objectives of this work are to acquire EOG signals and then use them for control applications based on Arduino microcontroller. In the presented work, EOG signals are easily collected and processed in real time with a hardware control device. With the proposed system, it is possible to help the disabled patients to perform their tasks in an easy way.

The rest of this paper is structured as follows. Section II presents the materials and methods. Section III presents the flowchart of the control system. Section IV covers the experimental setup. Section V presents the results and discussions. Finally, in Section VI, the conclusion and the future work are presented.

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II. MATERIALS AND METHODS

A. System Configuration

The acquisition of EOG signals in the HCI system based on Arduino microcontroller consists of hardware and software interfaces. The block diagram of the acquisition and control system using EOG, and Arduino microcontrollers is shown in Fig.1.

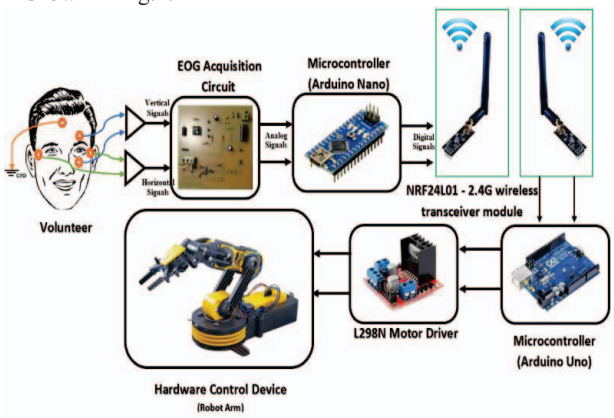


Fig. 1. Overview of the system configuration.

B. System Requirements

This system requires volunteers, Ag/Agcl electrodes sensors, EOG acquisition circuit, Arduino Nano-R3 microcontrollers, NRF24L01 - 2.4G wireless transceiver module for master and slave circuit, Arduino Uno microcontroller, L298N motor driver, robot arm, Laptop, Matlab program, Arduino program, Proteus 8 professional simulation program and IDL-600 ANALOG LAB unit for power supply.

C. System Needed and Challenges

There are many people with disabilities in their hands and legs. So, there is a bad need for systems to help the disabled perform their main tasks in their daily lives. One of the challenges we faced in this work was how to take the signal from the eyes of the volunteers. On other hand, the correct way to place the electrodes in the horizontal and vertical directions is another issue that deserves consideration. Another challenge is how to convert the analogue signal to a digital EOG signal. The most challenging part is how to take advantage of this digital data to control the robotic arm in a wireless manner to assist the handicapped.

D. EOG Acquisition circuit

Fig.2 and Fig.3 show the circuits of single-channel EOG signal acquisition using the AD620-IA and LM324N in simulation and printed circuit board, respectively.

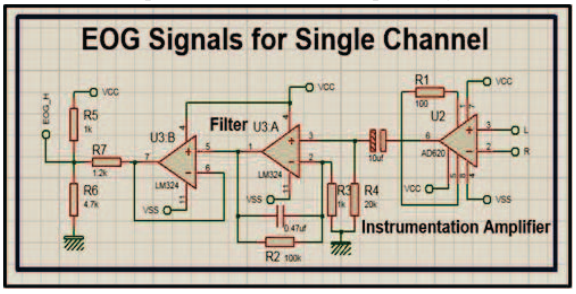


Fig. 2. EOG signal acquisition circuit in simulation.

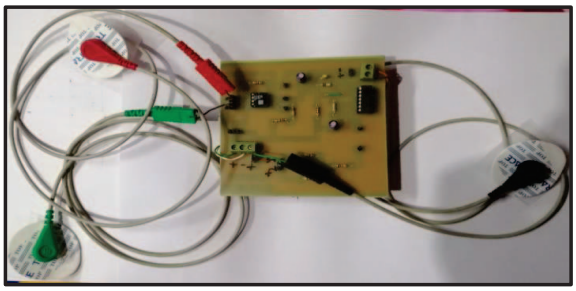


Fig. 3. EOG signal acquisition circuit in printed circuit board.

The gain of the AD620-IA depends on the gain resistance (Rg) in equation (1).

$$Gain = \frac{49.4k\Omega}{Rg} + 1 \quad (1)$$

where Rg=100 Ω

$$Gain = \frac{49.4k\Omega}{100} + 1 = 495$$

After the amplification process, a high-pass filter is used to remove the DC offset and low-frequency noise. The bandwidth of the EOG signals in the presented work is from 0.1 to 10 Hz. The cut-off frequency of the filter is determined by equation (2).

$$fc = \frac{1}{2\pi RC} \quad (2)$$

E. Volunteers and Electrode Placement

A volunteer is a person, who generates the signals in both horizontal and vertical directions such as close eye, open eye, lift, right, up and down. The electrodes are placed on the face as shown in Fig.4. Then, the signals are transmitted to the EOG circuit.

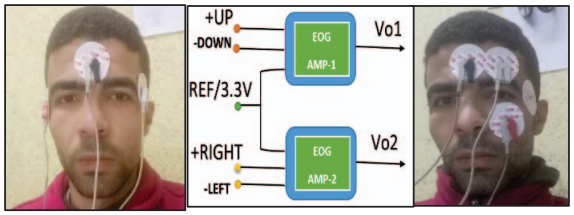


Fig. 4. Electrodes Placement and configuration.

The vertical and horizontal direction signals are obtained by the upper and lower eye movements. The reference electrode (G) is placed on the forehead.

III. FLOWCHART OF THE CONTROL SYSTEM

The volunteer sits on the chair and looks in the front direction. The EOG signals are acquired, when the volunteer moves his eye to the left, right, front, up, and down directions. The corresponding pictorial movement view of horizontal and vertical movements of the eye are shown in Fig. 5. The output EOG signal is converted from analog to digital form by using the Arduino microcontroller. This data ranges from 0 to 1023.

The values in red fonts in Fig.5 show the ranges of the horizontal and the vertical eye movements. The vertical direction is divided into three ranges: the lower ranges from 0 to 410, the middle ranges from 410 to 640, and the upper ranges from 640 to 1023.

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On other hand, the horizontal direction is divided into three ranges: the left ranges from 0 to 350, the middle ranges from 350 to 750, and the right range from 750 to 1023. Each range is equivalent to a fixed ranges from 0 to 255 for PWM of motor speed.

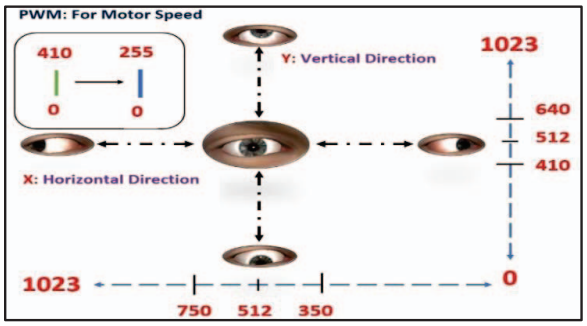


Fig. 5. Pictorial view for vertical and horizontal movement of the eye.

The flowchart as shown in Fig.6 presents the steps of the master (transmit) circuit and the slave (receive) circuit to control of the robotic arm using the presented system.

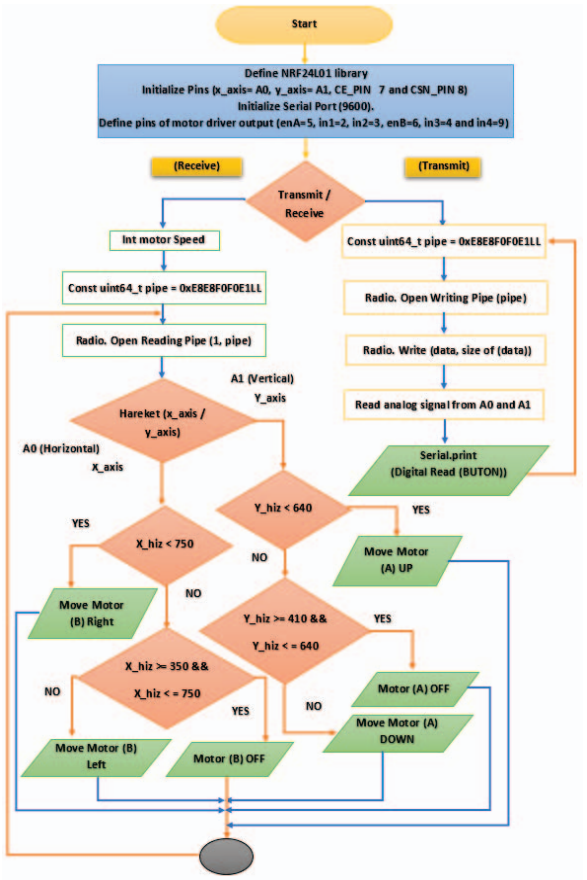


Fig. 6. The flowchart of the proposed system.

For the master process (transmitter), a library of NRF24L01- transmit is defined and the pins A0 and A1 for the horizontal x-axis and the vertical y-axis signals are initialized. The pin of the transmitter unit and transmission channel are initialized and the Serial.begin is set to 9600. The measured signals are obtained through the pins A0 and A1 of the Arduino. Then the signals are sent through the transmitter circuit.

For the slave process (receiver), a library of NRF24L01-receiver is defined and the motor terminals of the robotic arm are initialized. The signals are received by the slave circuit. These signals are sent to the motors that are connected to the robotic arm. The robotic arm is controlled by the ranges of each direction, vertically and horizontally. A map is made equivalent to each range for a movement in a certain direction with a digital range from 0 to 255 to control the motor speed as follows:

Motor Speed A = map (y_hiz, 640, 1023, 0, 255)

IV. Experimental Setup

A. Acquisition of EOG Signals

The experimental setup using AD620-IA and LM324N to acquire the signals for one channel of EOG circuit as shown in Fig.7.

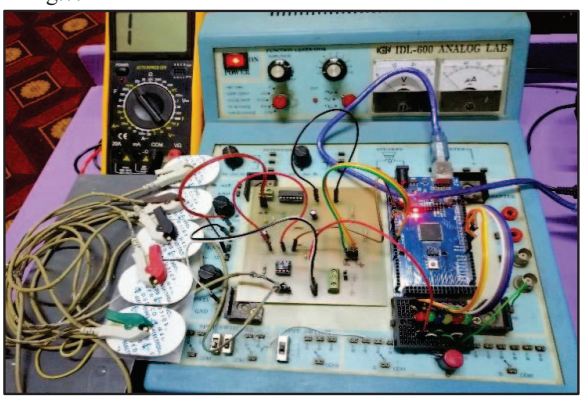


Fig. 7. Experimental Setup of EOG signals in printed circuit board.

In this experiment, the Arduino microcontroller unit is used to get data from eye movements. It interfaced with MATLAB-SIMULINK and Arduino program, as shown in Fig.8 and Fig.9.

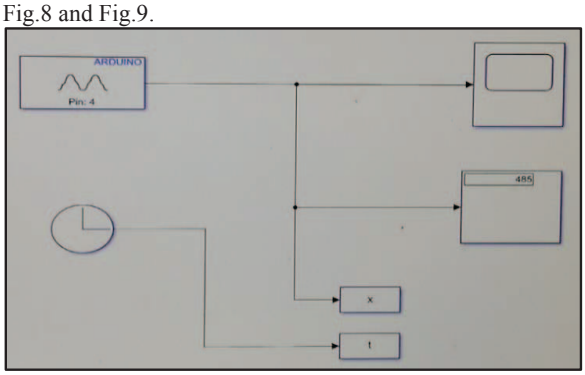


Fig. 8. MATLAB / SIMULINK for acquisition EOG signals.

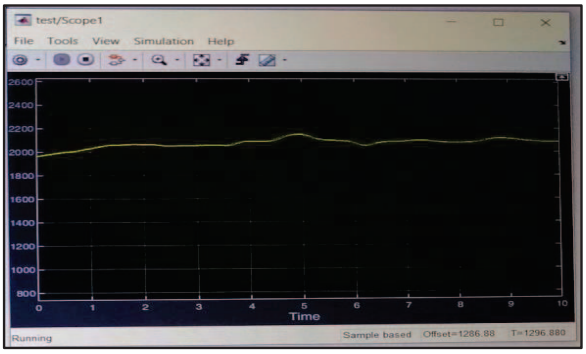


Fig. 9. Scope of EOG signals in MATLAB / SIMULINK.

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B. Control of Robot Arm Based on EOG Signals

A wireless robotic arm is used as the representative assistive device. The vertical and horizontal EOG signals of eye movements are used to control the robot arm for six fundamental directions. The overall view of the customized robotic arm and the detailed experimental design are shown in Fig.10. The experimental design consists of an EOG acquisition circuit on test board, a master circuit, a slave circuit, a Laptop, and an IDL-600 ANALOG LAB unit for power supply.

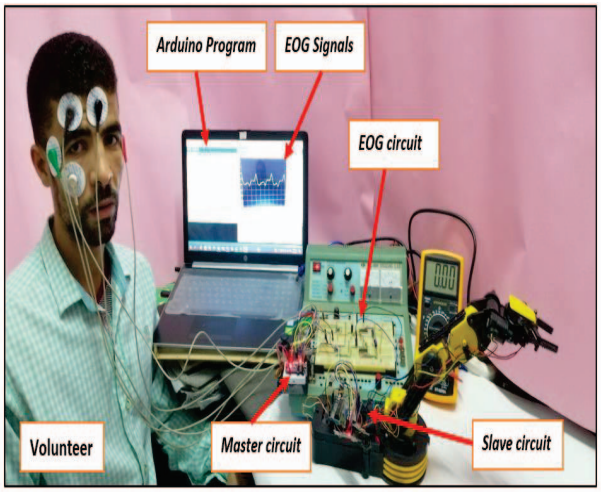


Fig. 10. Overall view of the customized robotic arm.

The master circuit consists of an Arduino Nano microcontroller, a button, a battery and an NRF24L01 - 2.4G wireless transceiver module to transmit the signals, as shown in Fig.11.

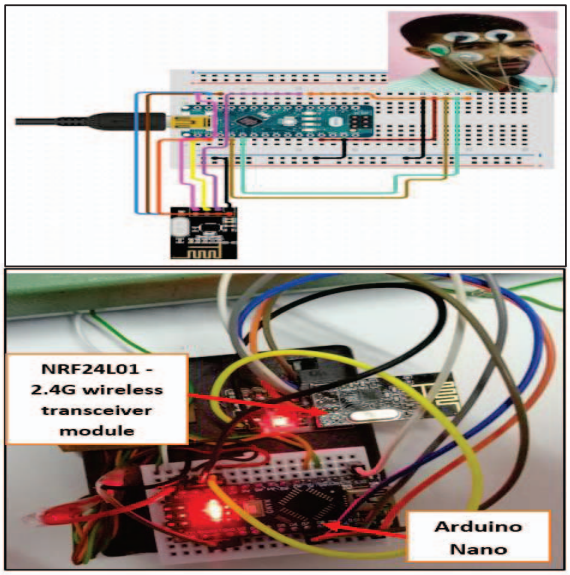


Fig. 11. NRF24L01 - 2.4G wireless transceiver module for master circuit.

The slave circuit consists of an Arduino Uno microcontroller, a motor driver circuit that is designed using L298N, a robot arm and an NRF24L01 - 2.4G wireless transceiver module to transmit the signals, as shown in Fig.12.

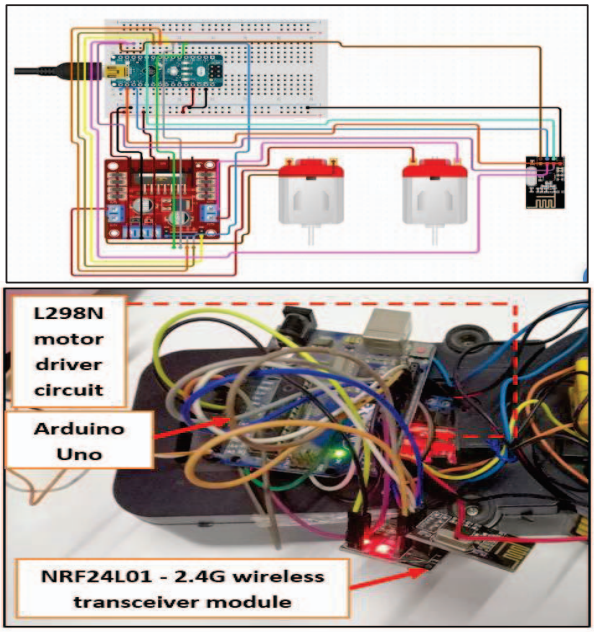


Fig. 12. NRF24L01 - 2.4G wireless transceiver module for slave circuit.

In the experimental design, the programs of the transmitter and the receiver circuits are uploaded. After that, the volunteer sits and begins to perform basic eye movements. These movements represent the horizontal and vertical direction movements. For example, to execute the forward, right and left direction movements, the EOG signals in the master circuit transmit the analog signals to the Arduino Nano microcontroller.

The signals are processed and converted to digital signals. A map is made equivalent to each range for a movement in a certain direction with a digital range from 0 to 255 to control the motor speed as follows:

$$x_hiz = map\ (x_hiz, 0, 350, 0, 255)$$

This map makes the range of levels from 0 to 350 equivalent to the range from 0 to 255 PWM for motor speed.

The signals are transmitted from the master circuit and received by the slave circuit. The signals are processed and transmitted to the L298N motor driver. The motor moves the robot arm through the movements acquired from the volunteer at a transmission rate of 9600. This process is repeated for transmitting and receiving all movements made by the volunteer through the system to control the robotic arm.

V. RUESULTS AND DISCUSSION

The EOG data are collected, and the results are analyzed to examine the effect of eye movements, such as close, open, right, left, up and down. The signals are obtained using MATLAB-SIMULINK and Arduino program software.

In the acquisition of EOG signals, two signals are observed: one without filtering as shown in Fig.13 and the other with filtering as shown in Fig.14 with a gain of 495 using MATLAB R2019.b. The AD620-IA has a resistor Rg=100Ω. The EOG signal is received, and the signals are amplified using an amplifier with a gain of 495.

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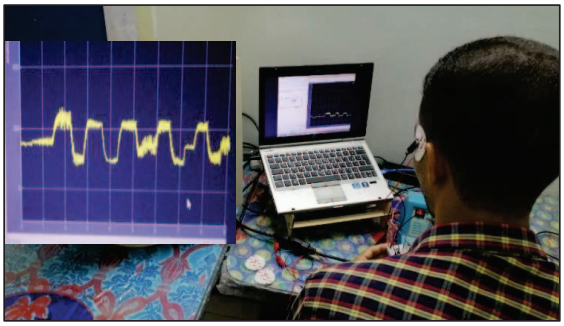


Fig. 13. Instrumentation amplifier output without filtering.

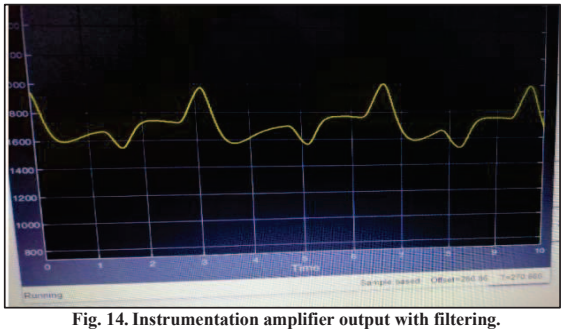


Fig. 14. Instrumentation amplifier output with filtering.

The raw EOG signal is filtered by an active high-pass filter. The waveforms show that the low-frequency component is almost removed. Positive peaks and negative peaks are observed, when the eye moves in right and left directions for horizontal movement. Similar results are obtained for vertical movement. Fig.15 and Fig.16 show the waveform characteristics of the instrumentation amplifier output with and without filtering.

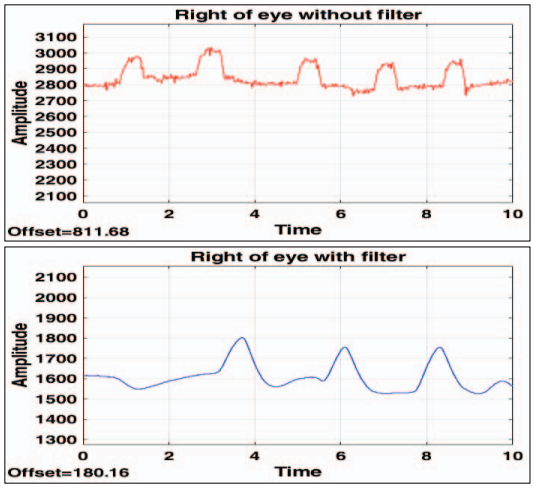


Fig. 15. Characteristics of right eye measured signals without and with filtering.

The signals are processed on the MTLAB program. They are converted to double format, which can be dealt with directly in the feature extraction and classification processes. The output EOG signals are converted from analog to digital through the MATLAB and Arduino programs. The digital data can be used in the program to make feature extraction and classification, which can be used in various applications for the handicapped.

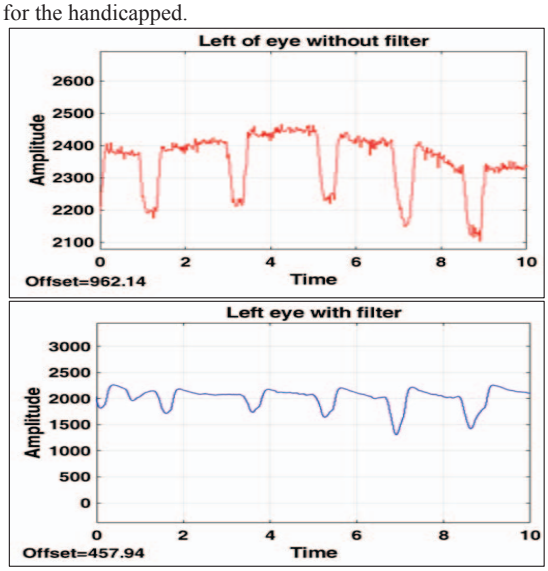


Fig. 16. Characteristics of left eye measured signals without and with filtering.

Table I shows data ranges of the output form for the EOG signals in the horizontal and vertical eye movement directions. The signals are converted from analog to digital forms. The sample values range from 0 to 1023. The samples are obtained from pins A0 and A1 in the microcontroller. From Table I, it is clear that the digital conversion outputs of horizontal and vertical data values change depending on the eye movements for every volunteer.

TABLE I. DIGITAL CONVERSION OUTPUTS OF EOG SIGNALS IN ARDUINO PROGRAM.

Eye movement directions (Ranges of data)			
Horizontal EOG signal data		Vertical EOG signal data	
Volunteer 1	*Volunteer 2*	*Volunteer 1*	*Volunteer 2*
990	637	0	1023
895	640	4	990
866	643	23	895
877	644	87	866
870	653	132	877
837	657	196	870
819	479	260	837
764	333	310	819
562	196	348	764
285	75	368	620
114	82	381	600
54	123	461	586
24	160	617	562
0	223	818	285
28	360	957	114
55	554	1020	24
85	761	1023	0
142	921	1013	21
213	1004	554	85
279	1023	897	142
357	995	853	270
990	637	0	1023

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The previous work used the EEG acquisition hardware for EOG signal acquisition and Bluetooth to transmit and receive the data. In the presented system in this paper, the EOG acquisition circuit and NRF24L01 - 2.4G wireless transceiver module have been used for master and slave circuits. This makes the performance and accuracy of the presented system really promising due to simplicity and reliability.

VI. CONCOLUSION AND FUTURE WORK

In this paper, EOG signals have been used to control a wireless robotic arm. Ag/AgCl electrodes were used to acquire the EOG signals. The signals are processed on a real-time basis, and subsequently classified for decision making with MATLAB and an Arduino based program. The NRF24L01 - 2.4G wireless transceiver module is used to transfer data from master to slave circuits. Arduino Nano-R3 is connected to the master circuit, which contains the volunteer, EOG circuit and microcontroller. The master circuit is connected to the USB port of the laptop to upload the code to the Arduino. At the receiver end, the Arduino Uno microcontroller is connected to the slave circuit, which contains an electronic module transceiver, with an L298N motor driver circuit mounted on a customized robotic arm. The performance and accuracy of the presented system are really promising due to simplicity and reliability. This trend can help the disabled patients in rehabilitation aids and other applications to improve their quality of life by simply moving their eyes via this presented system.

In the future work for EOG acquisition systems, the following points are recommended:

- An EOG acquisition system based on camera, Microsoft Kinect XBOX 360TM and ATmega AVR microcontroller with Lab view and Matlab should be introduced.
- PSL-iEOG2 (2-Channel EOG Module) should be used for rehabilitation purposes and other applications that help the handicapped to take decisions for better quality of life, by providing possible human interaction with a computer.

ACKNOWLEDGEMENT

The authors would like to thank Faculty of Technology and Education, Electrical Specialization and Beni-Suef University for the support.

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